
CS 421 --- Type Semantics Activity (Monotype Version)

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

The Rules

The Language

$L ::=$	$\lambda x.L$	abstractions
	$ \quad L \ L$	applications
	$ \quad \mathbf{let} \ x = L \ \mathbf{in} \ L$	Let expressions
	$ \quad \mathbf{if} \ L \ \mathbf{then} \ L \ \mathbf{else} \ L$	If expressions
	$ \quad E$	expressions
$E ::=$	x	variables
	$ \quad n$	integers
	$ \quad b$	booleans
	$ \quad E \oplus E$	integer operations
	$ \quad E \sim E$	integer comparisons
	$ \quad E \ \&\& \ E$	boolean and
	$ \quad E \ \ E$	boolean or

Reductions

Reduce the following programs according to the semantic rules given.

Problem 1)

$\{x:\text{Int}, y:\text{Int}\} \vdash \mathbf{if} \ x * y > 2 \ \mathbf{then} \ x \ \mathbf{else} \ y : \text{Int}$

Problem 2)

$\{x:\text{Int}, y:\text{Int}\} \vdash \mathbf{let} \ m = x * y \ \mathbf{in} \ m - x : \text{Int}$

Problem 3)

$\{\} \vdash (\lambda f. \lambda x. f \ x) (\lambda x. x) \ 10 : \text{Int}$

Make your own rules!

Problem 4)

Try to write the type rules for `HASKELL`'s `head` and `tail` functions.

Problem 5)

The logical rule for *Modus Ponens* looks like this:

$$\frac{A \rightarrow B \quad A}{B}$$

Is there a programming language equivalent to this?¹ Talk to a neighbor and see if you can find a semantic rule that mirrors this.

Problem 6) What happens when you try to type check this code? Try to derive α .

$\{y:\text{Int}, z:\text{String}\} \vdash (\lambda f.(fy, fz)) (\lambda x.x) : \alpha$

¹Hint, the answer is “yes”.